MILITARY SPECIFICATION

# PITOT STATIC TUBE, L-SHAPED, COMPENSATED, ELECTRICALLY HEATED, GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

## 1. SCOPE

1.1 Scope - This specification covers the general requirements for aerodynamically compensated, L-shaped, pitot static tubes.

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-D-1000/1 | Drawings, Engineering and Associated Data |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-F-5509 | Fittings, Flared Tube, Fluid Connection |
| MIL-C-5541 | Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys |
| MIL-C-5778 | Covers, Aircraft Components |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification for |

FSC 6610

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SPECIFICATIONS

Military (Continued)

| | |
|---|---|
| MIL-A-8625 | Anodic Coating, for Aluminum and Aluminum Alloys |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-781 | Reliability Tests, Exponential Distribution |
| MIL-STD-785 | Requirements for Reliability Program (for Systems and Equipments) |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-1247 | Identification of Pipe, Hose, and Tube Lines for Aircraft, Missiles, and Space Systems |
| MS 33586 | Metals, Definition of Dissimilar |
| MS 33656 | Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal |



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2.2 Availability of Documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

## 3. REQUIREMENTS

3.1 Detail Specifications - The individual item requirements shall be as specified herein and in accordance with the applicable slash sheet detail specifications.

3.2 Qualification - The pitot-static tubes furnished under this specification shall be products which have been tested and meet the quality assurance provisions specified herein and have been listed on or approved for listing on the applicable qualified products list at the time set for opening of bids.

3.3 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.2 Standard Parts - With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials - Materials that are not specifically designated herein shall be of the best commercial quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Non-magnetic Materials - Non-magnetic materials shall be used for all parts of the tube except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity.



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3.4.2 Protective Treatment - When materials are used in the construction of the pitot tube that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.3 Toxic and Corrosive Fumes - The materials of the tube shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.4.4 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.4.5 Nonferrous Materials - Nonferrous materials shall be used in all parts of the tube, except where ferrous materials are essential.

3.4.6 Fungus Resistance - Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.7 Dissimilar Metals - Dissimilar metals as defined in MS 33586 shall not be used in intimate contact with each other, unless protection against electrolytic corrosion is provided.

3.4.8 Components - The electronic and mechanical components of the tube shall be in accordance with MIL-E-5400.

3.5 Design and Construction - The design of the tube shall be such that the tube will sense pitot and static pressures within the accuracies specified in the detail specification. Fixed aerodynamic compensation shall be achieved by providing an appropriate contour on the nose or by a method whereby the accuracy required in the detail specification will be achieved over the specified speed range without the use of a Mach switch. The static pressure position error (in the region where the tube will be installed) is shown in the detail specification. The supplier shall investigate the effects of angle of attack and sideslip on the sensed pressures at this location. Based upon these considerations, a suitable compensating contour and suitable location of static ports shall be determined. The tube shall be installed on the aircraft as shown in the detail specification. When two



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tubes are utilized, provisions shall be made to prevent misinstallation of right and left hand units if they are not interchangeable. The tube dimensions shall conform to the detail specification and shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation and service.

3.5.1 Electrical Connector - The electrical connector shall be as specified in the detail specification.

3.5.2 Heater - The tube shall have one heater which shall be located in the head of the tube and in the strut. Power distribution through the heater during the deicing operation shall conform to the minimum wattage densities specified in the detail specification. The total overload wattage at any time shall not exceed 1,325w. Heater elements shall be placed as near as practicable to the heated surfaces. The heater shall be regulated automatically in such a manner that the power dissipated through the heater will be an inverse function of the heater element temperature. In still air at an ambient temperature between 20° and 30° C, the heater output shall not exceed one half the stabilized deicing power after 5 minutes of heater operation.

3.5.3 Pressure Lines - The pitot pressure exit line shall consist of 1/4 inch outside diameter tubing having a wall thickness of 0.022 inch. The line fitting shall conform to MIL-F-5509 and MS33656-4. The static pressure line shall consist of tubing no smaller than 1/4 inch outside diameter having a wall thickness of 0.022 inch. The static pressure line fitting shall conform to MIL-F-5509 and MS33656-5 or MS33656-6, as applicable. The minimum area of the pitot and static internal pressure lines shall be not less than that of equivalent 1/4 inch tubing.

3.5.4 Pitot Pressure Chamber - A pitot chamber shall be provided, the volume of which shall not exceed 4 cubic inches. The pressure lines shall be so connected to the plenum chamber that no moisture can enter the lines.

3.5.5 Static Pressure Chamber - The cross section of the static pressure chamber shall be not less than the cross section of 1/4 inch outer diameter tubing (0.022 inch wall thickness) over the entire length of the static pressure chamber.

3.5.6 Concentricity - The cylindrical portion of the tube body shall be concentric and uniform within 0.001 inch. There shall be no surface irregularities forward of, or within 1 inch aft of, the static pressure orifices.

3.5.7 Drainage - The tube shall be so designed that water cannot directly enter the pressure line under any service condition. One or two drain holes



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shall be provided in the pitot pressure chamber to clear any moisture that may collect. Drain holes shall be $0.029\ ^{+.002}_{-.001}$ inch in diameter. The drain holes shall be located in a heated area.

3.5.8 Screw Threads - The threads of all machine screws shall conform to MIL-S-7742.

3.5.9 Weight - The weight of the tube, including the electrical connector and mounting screws, shall not exceed that specified in the detail specification.

3.5.10 Finishes - Finishes and protective coatings that will crack, chip or scale during normal life or due to extremes of atmospheric conditions shall not be used. The outside of the tube shall have a durable dull finish acceptable to the procuring activity. The cylindrical portion of the tube shall have a machined surface of 32 microinches or better.

3.5.11 Protective Covers - The pitot and static pressure openings shall be protected by a full length cover. The pitot and static pressure line coupling threads shall be protected by slip-on covers of nonstrategic material. The design of all protective covers shall conform to MIL-C-5778 and shall be subject to the approval of the procuring activity.

3.5.12 Interchangeability and Replaceability - All tubes furnished under this specification having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Maximum commonality between airplane parts shall be a design objective. If parts are not common on each model, positive indexing shall be utilized to prevent incorrect installation. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000/1.

3.5.13 Electromagnetic Interference - The heating element of the tube shall be so designed that it will not be a source of electromagnetic interference. The requirements of MIL-STD-461 shall govern.

3.6 Performance - When installed at the location specified in the detail specification, the tube shall sense pitot and static pressures. When installed on the aircraft, the static pressure sensed by the tube shall be such that the residual errors will not exceed the limits as specified in the detail specification. The tube shall perform as specified when subjected to the environmental conditions specified herein.



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3.6.1 Primary Performance Characteristics - The tube shall be capable of meeting the following primary performance characteristics:

a. Nominal aircraft static pressure altitude error requirement - When the tube is installed on the aircraft at the specified location, the nominal or mean residual pressure error at a particular Mach number shall not exceed the values shown in Table I at any altitude or aircraft weight except between Mach 0.95 and 1.05.

TABLE I

| Flight Mach No. | $\Delta$ H (Feet) |
|---|---|
| 0.20 | 20 |
| 0.30 | 40 |
| 0.40 | 60 |
| 0.50 | 100 |
| 0.60 | 140 |
| 0.70 | 200 |
| 0.80 | 270 |
| 0.90 | 360 |
| 0.95 | 450 |
| 1.05 | 750 |
| 1.10 | 750 |
| 1.20 | 750 |
| 1.40 | 750 |
| 1.60 | 750 |
| 1.80 | 1000 |
| 2.00 | 1000 |

b. Variation allowance for nominal or mean residual pressure error - The combined error effect of reproducibility, installation, aircraft weight, altitude, aircraft configuration, and flight Mach number in level flight shall be within the tolerances specified in Table II.

TABLE II

| Flight Condition | Allowable* Tolerance, Ft |
|---|---|
| Power Approach | ±20 |
| Maximum Endurance Cruise | ±50 |
| Maximum Range Cruise | ±50 |
| Maximum Mach Number Obtainable | |
| Basic Engine (Continuous Power Operation) | ±50 |
| Military Power Operation | ±100 |

* Allowable tolerance may be defined as the variation around a nominal or mean.



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c. Static pressure reproducibility requirement between pitot-static tubes - The static pressure reproducibility of ±25 feet at maximum range cruising altitude and Mach number shall be substantiated by aerodynamic testing of each pitot-static tube. The test method with substantiating data shall be submitted to the procuring activity for approval.

d. Pitot and static pressure errors - The tube shall sense pitot and static pressures within the tolerance specified in the detail specification.

e. Deicing and anti-icing - The tube and strut heater shall remove and completely clear accumulated ice from heated areas when subjected to the environmental conditions specified in the detail specification.

3.6.2 Secondary Performance Characteristics - The tube shall be capable of meeting the following secondary performance characteristics:

a. Rated voltage - Operation from single phase power, 115V 400Hz, category "B" of MIL-STD-704. Normal voltage fluctuation shall have no effect on the unit.

b. Insulation - Voltage potential of 550 VAC 400 Hz between electrical connector pins and the tube body.

c. Pressure leakage - The tube shall withstand a pressure of 80 inches Hg for a period of 1 minute without leakage or permanent distortion.

d. Power consumption - At the end of a 5-minute period at rated voltage and still air conditions, the electrical power consumption shall not exceed one half the stabilized de-icing power of the tube.

e. Heater distortion - There shall be no permanent distortion of the tube as a result of extended heater operation.

f. Heater operation - Operation at rated voltage for a period of 10 hours.

3.6.3 Reliability - The tube shall be completely reliable in sensing pitot and static pressures throughout the operating life. Reliability of the tube heater and electrical circuit shall be such that the heater will satisfactorily perform its function throughout a life of 1000 hours under stringent environmental conditions. The contractor shall establish a reliability program in accordance with MIL-STD-781 and MIL-STD-785.



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3.6.4 Maintainability - The tube shall be a nonreparable item. The maintenance requirements shall include no more than periodic inspection of pitot and static pressure ports for damage, ingestion of foreign material and periodic checking of heater operation.

3.6.4.1 Operating Life - The tube shall have a minimum operating life of 1,000 hours under stringent heater operating conditions.

3.6.5 Environmental Requirements - The tube shall meet the performance requirements specified herein under the following conditions:

a. Vibration - Vibration including resonant and cycling conditions in each of the horizontal and vertical axes in accordance with MIL-STD-810, Method 514, Procedure I, Curve C, Parts 1, 2, and 3.

b. Pitot and static pressure error - Wind tunnel tests at Mach numbers and angles of attack as specified in the detail specification.

c. Salt fog - Exposure to salt sea atmosphere for a period of 48 hours.

d. High temperature exposure - Exposure to 400° C for a period of 48 hours.

e. Low temperature exposure and power consumption - Exposure to -65° C for 48 hours without damage; after 24 hours at this temperature, the tube shall withstand an initial power surge at rated voltage. The initial surge shall not exceed 1,325w.

f. Humidity - Relative humidity up to 100 percent.

3.7 Interface Requirements - Interface requirements are in regard to mounting on fuselage, pressure line connection, and mating electrical connector, and are specified in the detail specification.

3.8 Identification and Marking - The pitot-static tubes shall be marked for identification in accordance with MIL-N-18307. Identifying information shall be etched on each tube in a location that will not interfere with airflow over the tube or its operation. The pitot and static pressure lines shall be identified by the letters P and S, respectively, and shall be etched or stamped on the coupling in two places (both sides). Pressure lines shall be color-marked in accordance with MIL-STD-1247.

3.9 Storage - Design and packaging shall be such that storage will not affect tube performance.



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3.10 Workmanship - The tube shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies and freedom of parts from burrs and sharp edges. Static pressure ports must have sharp edges with no burrs.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - The inspection of the tubes shall be classified as follows:

a. Qualification inspection: Qualification inspection consists of examinations and tests performed on sample tubes submitted for approval as a qualified product.

b. Quality conformance inspection: Quality conformance inspection consists of examinations and tests performed on tubes manufactured and submitted for acceptance under contract.

4.2.1 Qualification Inspection - The qualification inspection of the pitot-static tube shall consist of all the examinations and tests of this specification performed in the order specified under paragraphs 4.5, 4.6, and 4.7 except for 4.7.1 and shall include the reliability test for the Reliability Qualification Phase.

4.2.1.1 Qualification Inspection Sample - Qualification inspection samples shall consist of three tubes manufactured in accordance with this specification. The tubes submitted for qualification inspection shall have been previously subjected only to the inspections of 4.5. Samples and data shall be forwarded at the supplier's expense to the laboratory designated in the Letter of Authorization.



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4.2.1.1.1 Qualification Inspection Sample Identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
Pitot Static Tube Type _____
Submitted by (Manufacturer's name, date)
for Qualification Inspection in accordance
with Specification MIL-P- under
authorization (reference letter authorizing tests).

4.2.2 Quality Conformance Inspection - The quality conformance inspection shall consist of the Individual Inspection, the Sampling Plans, and the Reliability Assurance Tests of this specification. The supplier shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling Plan B inspections shall be conducted as directed by the procuring activity. Quality conformance inspection shall be under the supervision of the Government quality control representative. The supplier shall furnish test reports showing quantitative results for all tests required by this specification (including detailed results of the examination of product test with respect to meeting each applicable specification requirement) signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.3 Individual Inspection - Each pitot-static tube submitted for acceptance shall be subjected to the individual tests to determine compliance with the requirements of material, workmanship and operational adequacy. As a minimum, each tube accepted shall have passed the following inspections:

a. Examination of Product

b. Insulation

c. Power Consumption

d. Pressure Leakage

e. Static Pressure Reproducibility

4.4 Sampling Plans - The sampling plans shall consist of Sampling Plan A, Sampling Plan B, and the Reliability Sampling Phase tests. The inspection samples which have been subjected to Sampling Plan A inspection



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shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling Plan B inspection shall not be delivered on contract as usable production equipment.

4.4.1 Sampling Plan A Sample Selection - Sampling Plan A samples shall be selected at random in accordance with the following schedule:

| Quantity Offered for acceptance | Quantity to be Selected for Inspection |
|---|---|
| First 15 | 1 (Zero when Sampling Plan B is invoked) |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| Each additional 200 or fraction thereof | 1 |

When a tube failure occurs, no tubes from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each fifteen from those still on hand, and proceed with sampling as indicated above when the next quantity is offered for acceptance.

4.4.1.1 Sampling Plan A Inspection - The tubes submitted for Sampling Plan A inspection shall have been previously subjected only to the Individual tests. Each sample selected for Sampling Plan A inspection shall be subjected to the following tests:

a. Vibration (4.6.1)

b. Heater Operation (4.6.2)

4.4.2 Sampling Plan B Instructions - Three samples shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise directed, these samples shall be forwarded at the contractor's expense to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Pitot Static Tube Type _____
Submitted by (Manufacturer's name, date) for
production acceptance Sampling Plan B inspection
of MIL-P- with Contract No. _____
Manufacturer's part number _____



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4.4.2.1 Sampling Plan B Inspection-Each sample selected for Sampling Plan B testing shall be subjected to the following tests:

a. Sampling Plan A Tests

b. Static and Pitot Pressure Error

c. De-icing and Anti-icing

d. Salt Fog

e. High Temperature Resistance

f. Low Temperature Resistance and Power Comsumption

g. Endurance Cycling

h. Humidity

i. Fungus

j. Radiographic Inspection

4.4.2.2 Sampling Plan B Approval- Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4.4.3 Reliability Assurance Tests- Reliability Assurance Tests shall consist of the Qualification Phase and Sampling Phase Tests. The tubes selected for the Reliability Assurance Tests shall first have passed the Individual tests.

4.4.3.1 Reliability Qualification Phase- Prior to the acceptance of the tubes under the contract or order, between 3 and 6 tubes shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests" using the duty cycle specified herein. Accept-reject criteria for Test Plan III shall be used. The tubes shall demonstrate a specified MTBF of 1,000 hours.

4.4.3.1.1 Duty Cycle - The tubes shall be subjected to the following environmental conditions during this test. The heater shall operate for 1 hour followed by 15 minutes with power off. The tubes shall be subjected to 200 cycles as specified in column B.



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| A | B | C |
|---|---|---|
| Percent of Test Time | Power on Time (Hrs.) | Environmental Conditions |
| 10 | 20 | Still air at -54°C |
| 20 | 40 | Still air at -25°C |
| 50 | 100 | Simulated airflow not to exceed 50 knots at 20°C to 30°C while being subjected to vertical axis vibrations, perpendicular to the tube centerline, in accordance with MIL-STD-810, Method 514, Procedure I. The tube shall be mounted by its base to simulate installation on the aircraft. |
| 20 | 40 | Still air at 20°C to 30°C with vibration as specified above. |

4.4.3.1.2 Test Time and Failures - The tubes shall be tested until an acceptance level is attained. The Pressure Leakage Test specified in 4.5.4 shall be repeated on each unit at maximum intervals of 100 hours. At maximum intevals of 50 hours, each tube shall be subjected to the Insulation Test specified in 4.5.2, sea level pressure only. The resistance of the heating element shall be measured and recorded. The tube shall then be subjected to and shall meet the requirements of the Static Pressure Reproducibility Test at maximum intervals of 100 hours.

4.4.3.2 Reliability Sampling Phase Test - Samples of the tube shall be tested per subparagraphs a, b, g, and j of paragraph 4.4.2.1.

4.4.3.2.1 Lot Size for Sampling Phase - The lot size shall not be less than 50 nor more than 125 tubes. One sample from each lot, selected at random, shall be tested. This test may utilize a Sampling Plan A sample. At any time that the test results show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc.

4.4.3.3 Disposition of Samples - Any sample tested may be delivered on contract provided the sample is in "good as new" condition or has been refurbished and meets the Individual Inspection Tests, and is representative of production units currently being accepted.

4.4.4 Test Conditions -



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4.4.4.1 Standard Conditions - Unless otherwise specified, the tube shall be inspected under the following conditions:

a. Input power - 115 V single phase, 400 Hz or 115 V single phase, 60 Hz commercial frequency.

b. Atmospheric pressure - approximately 29.92 inches Hg.

c. Room temperature - approximately 25°C.

d. Humidity - room ambient up to 90% relative humidity.

4.5 Individual Tests -

4.5.1 Examination of Product - Each tube shall be carefully examined externally to determine conformance with the applicable drawings and with each of the requirements of this specification and the detail specification capable of being inspected in this manner.

4.5.2 Insulation - The tube shall be subjected to a high-potential test at sea level pressure with 550 VAC commercial frequency applied between each terminal and the tube case for a period of 5 seconds. The electrical impedance during this test shall be not less than 2 megohms. Upon completion of testing, there shall be no damage to the wiring or insulation. The tube shall then be placed in a chamber evacuated to a pressure equal to an altitude of 48,000 feet and after 2 hours at this pressure, 550 VAC shall be applied between each terminal and the case for 5 seconds. No damage to the wiring or insulation shall result.

4.5.3 Power Consumption - At an ambient temperature of between 20°C and 30°C, rated voltage shall be applied through a 10-amp fuse to the terminals of the tube for a period of 5 minutes. After this 5-minute period, the power shall be measured and shall not exceed one half the stabilized de-icing power.

4.5.4 Pressure Leakage - With the pitot and static pressure orifices and drain holes sealed, a pressure of 80.0 inches Hg shall be applied to each of the pressure lines. With the source of pressure isolated, the trapped pressure drop after 1 minute shall not exceed 0.03 inches Hg.

4.5.5 Static Pressure Reproducibility - The tube shall be aligned within $\pm 0.5^{\circ}$ of a calibrated air flow of 85 KCAS. The tube static pressure output shall be recorded. Upon comparison of the data obtained from the test with that of a calibrated standard, the difference shall not be greater than $\pm 0.002\ q_c$.



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4.6 Sampling Plan A Tests -

4.6.1 Vibration - The tube shall be mounted by its base to simulate installation on the aircraft and subjected to vertical and horizontal vibrations perpendicular to the tube centerline in accordance with MIL-STD-810, Method 514, Procedure I, Curve C, Parts 1, 2, and 3. Pressure line connections shall simulate installation in aircraft. Rated voltage shall be applied to the heater during the entire test. No leaks or failures shall result from his test.

4.6.2 Heater Operation - At an ambient temperature of between 20°C and 30°C, rated voltage shall be applied to the heating element of the tube for an uninterrupted period of 60 minutes. Proper operation of the tube shall not be affected. The tube body shall be checked dimensionally to determine that no permanent distortion or warpage has resulted due to heater operation. The Pressure Leakage, Insulation, and Static Pressure Reproducibility Test shall then be repeated after which the tube body shall again be checked dimensionally to determine that no permanent distortion or warpage has resulted.

4.7 Sampling Plan B Tests -

4.7.1 Static and Pitot Pressure Error - The tube shall be mounted in a wind tunnel and tested to determine static and pitot pressure errors. The tube errors shall be determined by comparing the pressures sensed by the tube with actual tunnel pressures. The tube shall be tested at Mach numbers and angles of attack as specified in the detail specification. The pressure sensed by the static ports shall be such that when the tube is installed on the aircraft, the static pressure position errors will be corrected. The residual static pressure errors shall not deviate from the nominal curve shown in the detail specification by more than $\pm 0.01\ q_c$ to Mach 0.7 nor more than $\pm 0.005\ q_c$ above Mach 0.7. Static pressure position error at the mounting location is shown in the detail specification. The pitot pressure sensed by the tube at all angles of attack shall not differ by more than $0.005 + .0005\ \alpha$ (qc) from the actual pitot pressure in the tunnel. The supplier shall demonstrate a static pressure reproducibility of $\pm 0.002\ q_c$ by aerodynamic testing of each production pitot-static tube. The residual error of the tube shall be determined by flight tests which will be conducted by the procuring activity.

4.7.2 De-icing and Anti-icing - The tube shall be tested in an icing wind tunnel at an indicated tunnel speed of 350 ±25 knots. The ambient temperature for each test shall be -30° +5°C. The water droplet size shall be regulated to provide a liquid water content of 1.25 ±0.25 gram per cubic meter. Test runs shall be made at 0° and 15° angles of attack. The procedure for each test shall be as follows: Ice shall be allowed to form on the tube until either (a) the total pressure



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opening has been sealed, or (b) the ice cap has extended 1/2 inch in front of the tube lip. Power shall then be applied to the heater at rated voltage. The total time required to achieve correct pressure readings shall not exceed 1 minute and to remove all accumulated ice shall not exceed 1 minute and 30 seconds. The icing test shall be continued for 15 minutes after the ice cap is removed and ice shall not collect on the tube to such extent as to affect the pitot and static pressure.

4.7.3 Salt Fog - The tube shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509, Procedure I. At the end of the exposure period, the tube shall be subjected to and shall meet the requirements of the individual tests.

4.7.4 High Temperature Resistance - The tube shall be subjected to a temperature of 400°C for a period of 48 hours. At the end of this time, there shall be no evidence of damage. At the conclusion of the exposure period, the tube shall be subjected to and shall meet the requirements of the individual tests.

4.7.5 Low Temperature Resistance and Power Consumption - The tube shall be subjected to a temperature of -65°C for a period of 48 hours. After 24 hours at low temperature, rated voltage shall be applied to the tube through a 10 amp circuit breaker. The initial power surge shall not exceed 1,000w and shall not be of such a duration as to render the circuit breaker inoperative. After 5 seconds, the power consumption shall not exceed 750w and after 20 seconds shall not exceed 550w. Power shall then be removed. At the conclusion of the exposure period, the tube shall be subjected to and shall meet the requirements of the individual tests.

4.7.6 Endurance Cycling - At an ambient temperature between 20° and 30°C, the tube shall undergo an endurance cycling test in which rated voltage is applied to the tube for an uninterrupted period of 10 hours and then removed for 1 hour. The test shall be repeated for 30 consecutive cycles. There shall be no damage which will affect corrosion resistance or proper operation of the tube. At the conclusion of this test, the tube body shall be checked dimensionally to determine that no permanent distortion or warpage has occurred due to heater operation. The tube shall then be subjected to and shall meet the requirements of the Insulation, Pressure Leakage and Static Pressure Reproducibility Tests.

4.7.7 Humidity - The tube shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I. There shall be no detrimental effect on heater operation or deterioration of the tube body. At the completion of this test, the tube shall be subjected to and shall meet the requirements of the Insulation test.



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4.7.8 Fungus - The tube shall meet the requirements of a fungus test conducted in accordance with MIL-STD-810, Method 508, Procedure 1. There shall be no evidence of fungus growth on any part of the tube.

4.7.9 Radiographic Inspection - If the tube is fabricated in such a manner that a weld exists in the tube shell in the area near the intersection with the mounting strut, an X-ray inspection shall be conducted of the weld area. A minimum of three views displaced approximately 120° shall be taken, each at an angle to best view the weld penetration. Failure of the completed weld to extend around the complete circumference of the tube or detection of any gap or lack of fusion between weld material and base metal shall constitute failure of this test.

4.8 Rejection and Retest - Pitot-static tubes which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector. Failure of samples representing a lot shall be cause for rejection of the lot.

4.9 Flight Test Program - The program will consist of a flight test(s) of pitot-static tube(s) on the appropriate aircraft as noted in the detail specification. The static and pitot pressure errors shall be measured to determine compliance with the requirements of the detail specification. The flight tests will be conducted by, and shall be the responsibility of, the procuring activity. The data obtained shall be binding on the contractor.

4.10 Test Procedures - The procedures used for conducting quality conformance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirement of this specification or the contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures including additional tests when deemed necessary.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging - The pitot-static tube shall be packaged in accordance with MIL-STD-794, Level A or C as specified in the contract or order. For level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method 1A-8.

5.2 Packing- The pitot-static tube shall be packed in accordance with MIL-STD-794, Level A,B, or C, as specified in the contract or order.

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794.



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6. NOTES

6.1 Intended use - The pitot-static tubes covered by this specification are intended for use on various aircraft to supply pitot and static pressures for air data flight instruments and control equipment.

6.2 Ordering data - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether Sampling Plan B is to be invoked and disposition of samples after tests.

c. Type of tube desired (see applicable detail specification).

d. Whether Reliability Qualification Phase Tests are required (see paragraph 4.4.3.1).

e. Whether the Flight Test Program is to be omitted (see paragraph 4.9).

f. Levels of packaging and packing desired (see paragraphs 5.1 and 5.2).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is directed to Provisions Governing Qualification, SD-6, and they are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.4 Precedence of Documents - When the requirement of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Preparing Activity:
Navy - AS
Project No. 6610-N028



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R094

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-P-81655(AS) PITOT STATIC TUBE, L-SHAPED, COMPENSATED, ELECTRICALLY HEATED,

ORGANIZATION *(Of submitter)* GENERAL SPECIFICATION FOR | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20360

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